We assume that our initial population-distribution is symmetric and unimodal. Though we recognize that such an assumption may not be true in general in nature, for the time being we make this assumption for mathematical simplicity. Applying an increasing growth-function to a symmetric and unimodal distribution will result in a symmetric and unimodal population-distribution after growth. We also assume that the dispersal kernel is unimodal. Thus, we are taking the convolution of two symmetric, unimodal distributions, which will also be symmetric and unimodal (Wintner 1938; Dharmadhikari and Joag-Dev 1988; Purkayastha 1998). Many dispersal kernels satisfy the conditions of being symmetric and unimodal, including the popular Gaussian, Laplace, and Cauchy kernels. Using these assumptions, the population distributions given by the model will always be symmetric and unimodal, with the peak (population) density occurring at the center of the patch at $x = 0$ and the minimum density occurring at the edges of the patch, at $x = \pm L/2$.

We have constructed the simplest possible case for the population distributions of our IDE model using the above assumptions. This baseline case now allows us to explore the practicality of the block-pulse IDE and to identify and generalize any analytic patterns that emerge in the dynamics of our block-pulse models.

With these assumptions in hand, we consider an m-step block-pulse IDE. This IDE, with its growth function given by a block-pulse series, may be written as

$$\begin{aligned} n_{t+1}(x) &= \int_{-L/2}^{L/2} k(x-y) \sum_{i=1}^{m} g_i \phi_i [n_t(y)] \, dy \\ &= \sum_{i=1}^{m} g_i \int_{-L/2}^{L/2} k(x-y) \phi_i [n_t(y)] \, dy. \end{aligned} \tag{8}$$

As mentioned in Sect. 2.2, the coefficients g_i are different growth levels. As we have assumed our growth function to be strictly increasing, we have $g_i < g_{i+1}$ for $i = 1, 2, \ldots, m-1$.

Recall that each block-pulse function ϕ_i is nonzero only over a limited spatial region where it is equal to one. We may expand Eq. (8) into a sum of at most $2m-1$ integrals, where each integral is defined only over the spatial region(s) where $\phi_i \neq 0$ for each i. The boundaries of each region, $x = \pm r_{i,t}$, are the spatial thresholds where the population threshold is reached, $n_t(\pm r_{i,t}) = ih$. After expanding Eq. (8), using a change of variables $u = x - y$, and rewriting the resulting integrals in terms of cumulative distribution functions, we obtain the simple expression

$$\begin{aligned} n_{t+1}(x) = {} & g_1 [F(x + L/2) - F(x - L/2)] \\ & + (g_2 - g_1) \left[F\left(x + r_{1,t}\right) - F\left(x - r_{1,t}\right) \right] + \cdots \\ & + (g_m - g_{m-1}) \left[F\left(x + r_{m-1,t}\right) - F\left(x - r_{m-1,t}\right) \right], \end{aligned} \tag{9}$$

or

$$\begin{aligned} n_{t+1}(x) = {} & g_1 [F(x + L/2) - F(x - L/2)] \\ & + \sum_{i=1}^{m-1} (g_{i+1} - g_i) \left[F\left(x + r_{i,t}\right) - F\left(x - r_{i,t}\right) \right], \end{aligned} \tag{10}$$

where $F(x)$ is the cumulative distribution function (CDF) of a probability density function $k(z)$, defined as

$$F(x) = \int_{-\infty}^{x} k(z)\, dz. \tag{11}$$

We note that reducing our IDE model to Eqs. (9) or (10) amounts to approximating a Riemann–Stieltjes integral with a Riemann–Stieltjes sum. Due to Arzelà's (or Osgood's) bounded convergence theorem (Hildebrandt 1963; Luxemburg 1971; Apostol 1974; Monteiro et al. 2016, 2019), this series has nice convergence properties as $m \to \infty$.

The symmetry of the spatial thresholds $x = \pm r_{i,t}$ occurs because we have assumed our population distributions to be symmetric and unimodal. Thus, each spatial threshold where the population passes through a density threshold will occur twice in a symmetric fashion, if it occurs at all. We have $r_{i,t} > r_{i+1,t}$, with the t subscript denoting the time dependence of the spatial thresholds.

For this general example, we have assumed that the population passes through all density thresholds inside of the patch $[-L/2, L/2]$ and sees all m growth levels. Note that the population may fail to pass through a density threshold or may pass through it at a spatial threshold outside of the patch, in which case Eq. (10) has fewer terms.

We now have a model that is expressed entirely in terms of cumulative distribution functions, so that we have explicit analytic expressions for the entire spatial population-distribution in the next generation. This makes the block-pulse IDE analytically and numerically approachable, and the CDF form of the IDE is immensely useful. We now turn to a formal analysis of one-, two-, and three-step block-pulse IDEs.

4 Analytical results

As seen above, the block-pulse IDE may be rewritten into a remarkably simple form. Given the assumptions laid out in Sect. 3, block-pulse IDEs are a class of models with much more analytical tractability than most IDE models. We will now explore explicit equilibrium solutions and their regions of validity in parameter space, stability of equilibria, and bifurcations in one-, two-, and three-step block-pulse IDEs. In order to facilitate analysis, we will fix some of the block-pulse coefficients g_i to take on specific values. In later sections, where we apply the block-pulse IDE to particular growth functions, these growth levels are all explicitly given by Eq. (4) and the parameters of the original growth function.

Throughout the rest of this paper, we will illustrate examples using the Laplace kernel and its CDF, though the general trends we observe hold for other dispersal kernels satisfying our assumptions as well (see Sect. 6.2). The probability density function for the Laplace distribution is

$$k(z) = \frac{1}{2}\alpha e^{-\alpha|z|}, \tag{12}$$